SCSI 1013: Discrete Structure

CHAPTER 4

GRAPH THEORY

Trails, Paths & Circuits

Term and Description

- A walk from v to w is a finite alternating sequence of adjacent vertices and edges of G. Thus a walk has the form

$$(v_0, e_1, v_1, e_2, v_2, \ldots\ldots, v_{n-1}, e_n, v_n)$$

where the v's represent vertices, the e's represent edges, $v = v_0$, $w = v_n$, and for i = 1,2,...,n. v_{i-1} and v_i are the endpoints of e_i.

- A trivial walk from v to w consist of the single vertex v.

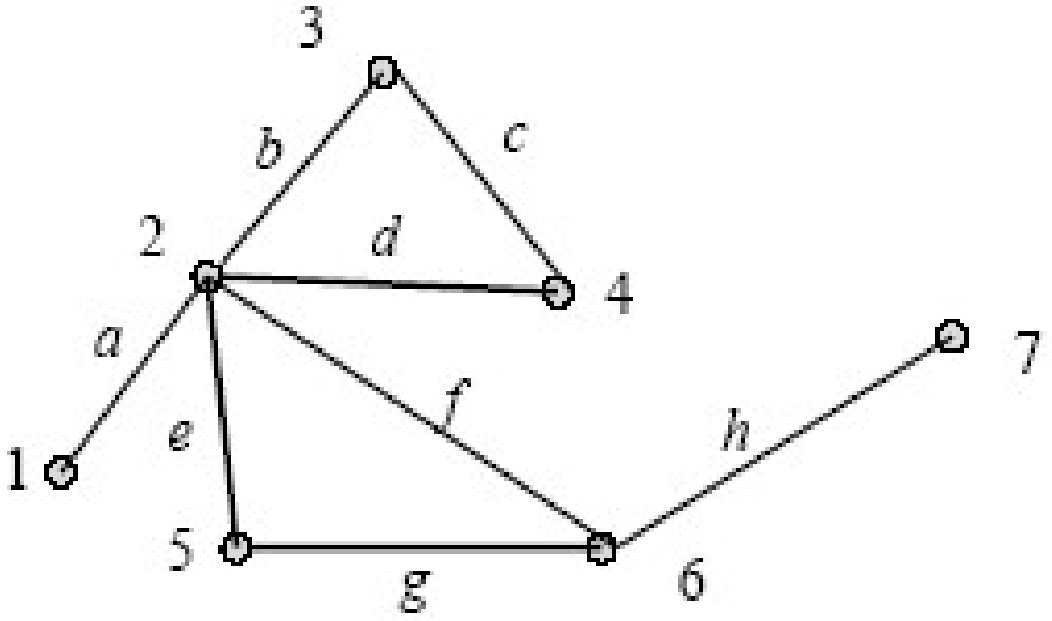
- The length of a walk is the number of edges it has.

Term and Description (cont.)

- A trail from v to w is a walk from v to w that does not contain a repeated edge.
- A path from v to w is a trail from v to w that does not contain a repeated vertex.
- A closed walk is a walk that start and ends at the same vertex.
- A circuit/cycle is a closed walk that contains at least one edge and does not contain a repeated edge.
- A simple circuit is a circuit that does not have any other repeated vertex except the first and the last.

Example 1 – Trail & Path

- (1, a, 2, b, 3, c, 4, d, 2, e, 5) is a trail.
- (6, g, 5, e, 2, d, 4) is a path.



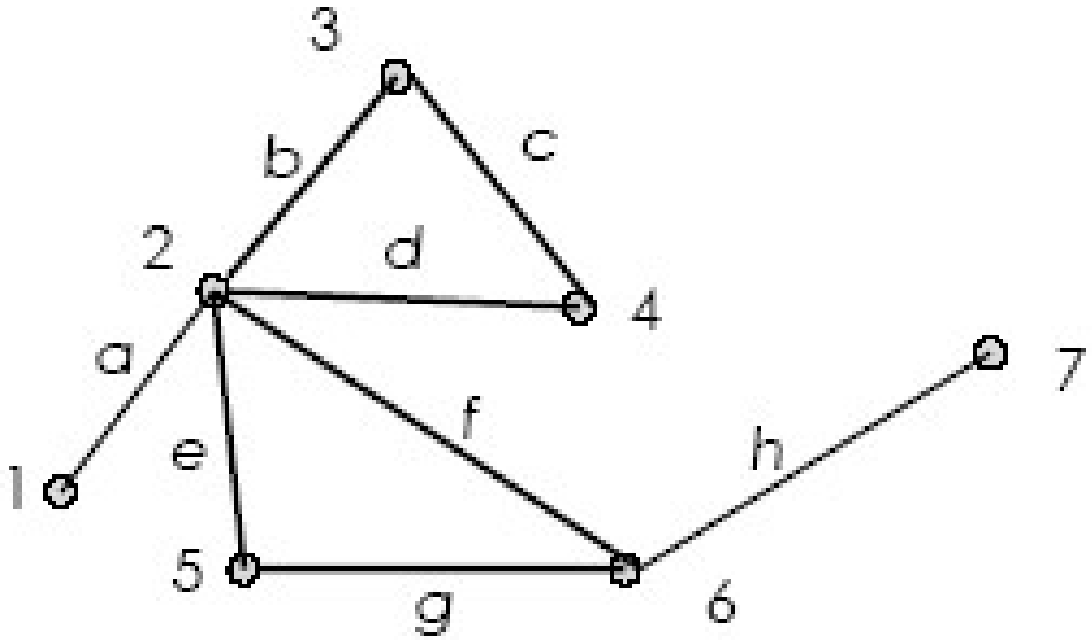
Note:

Trail: No repeated edge (can repeat vertex).

Path: No repeated vertex and edge.

Example 2 – Cycle/circuit

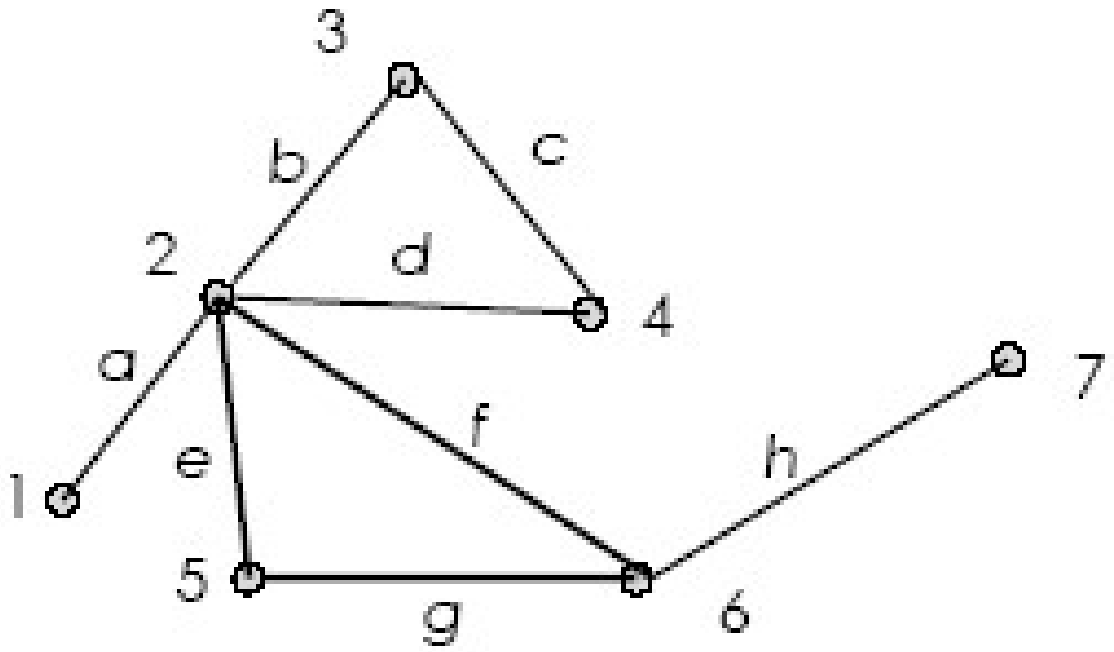
- (2, f, 6, g, 5, e, 2, d, 4, c, 3, b, 2) is a cycle.



Note: cycle –> start and end at same vertex, no repeated edge.

Example 3 – Simple Cycle

- (5, g, 6, f, 2, e, 5) is a simple cycle.

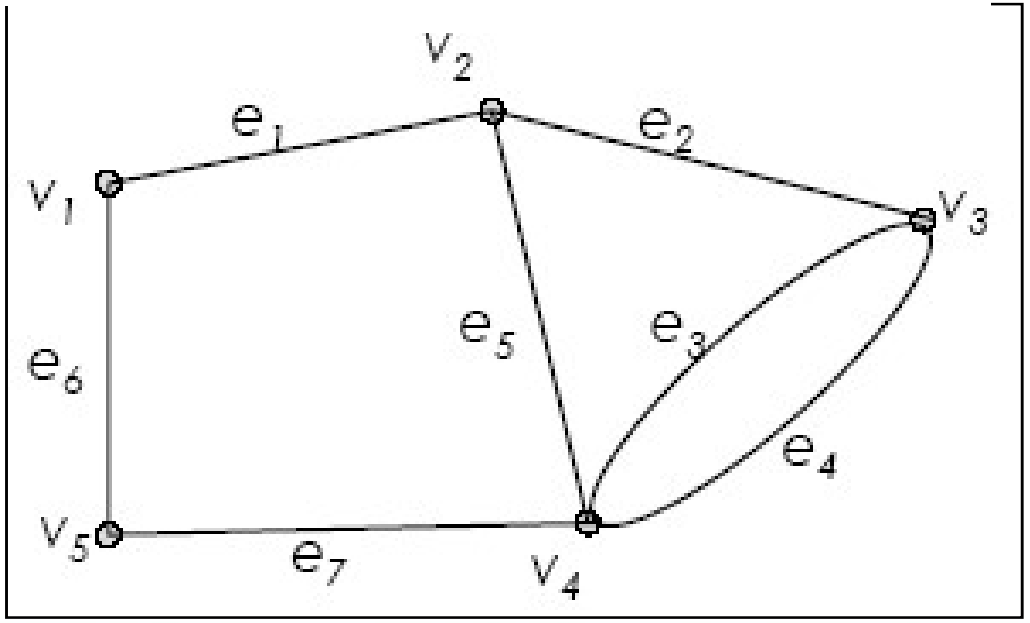


Note: Simple cycle –> start and end at same vertex, no repeated edge or vertex except for the start and end vertex.

Example 4

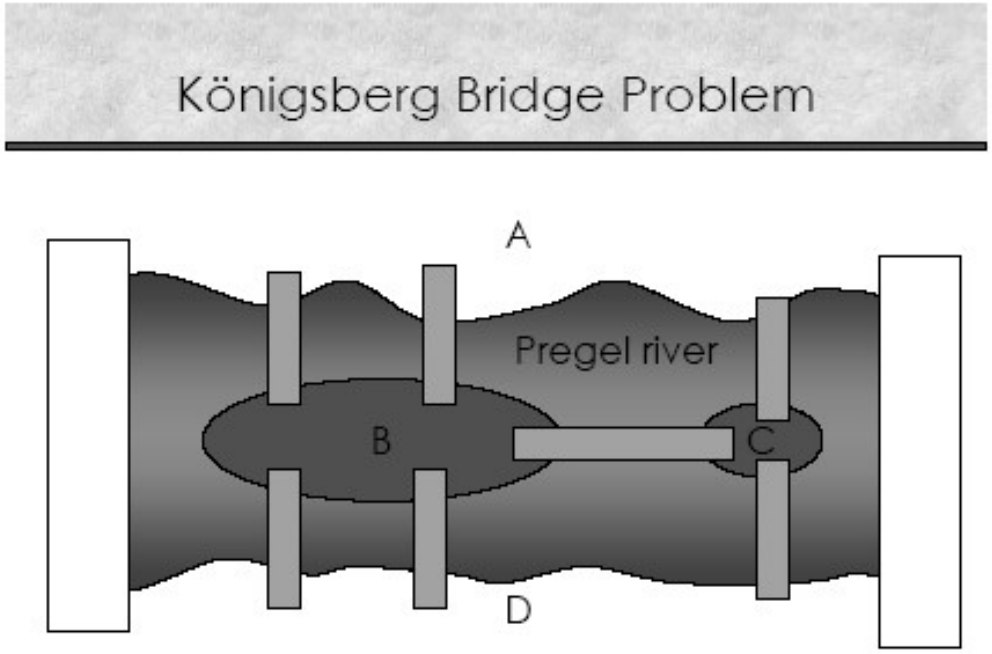
Tell whether the following is either a walk, trail, path, cycle, simple cycle, closed walk or none of these.

- (v_1, e_1, v_2) Path
- $(v_2, e_2, v_3, e_3, v_4, e_4, v_3)$ Trail
- $(v_4, e_7, v_5, e_6, v_1, e_1, v_2, e_2, v_3, e_3, v_4)$ Simple cycle
- $(v_4, e_4, v_3, e_3, v_4, e_5, v_2, e_1, v_1, e_6, v_5, e_7, v_4)$ Cycle



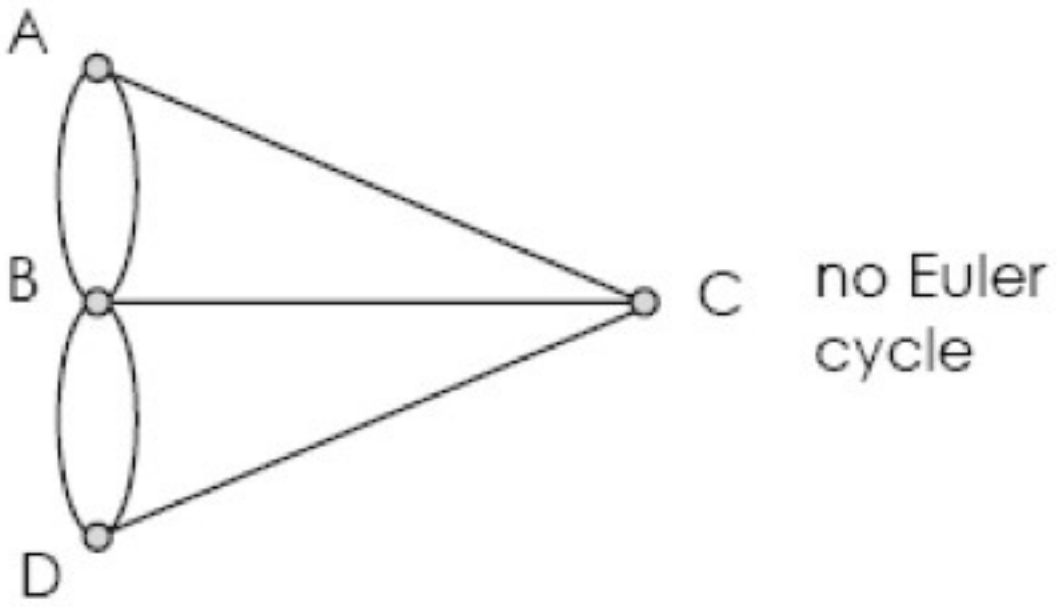
Euler Path & Circuit

The town of Konigsberg in Prussia (now Kaliningrad in Russia) was built at a point where two branches of the Pregel River came together. It consisted of an island and some land along the river banks. These were connected by seven bridges as shown in figure below:

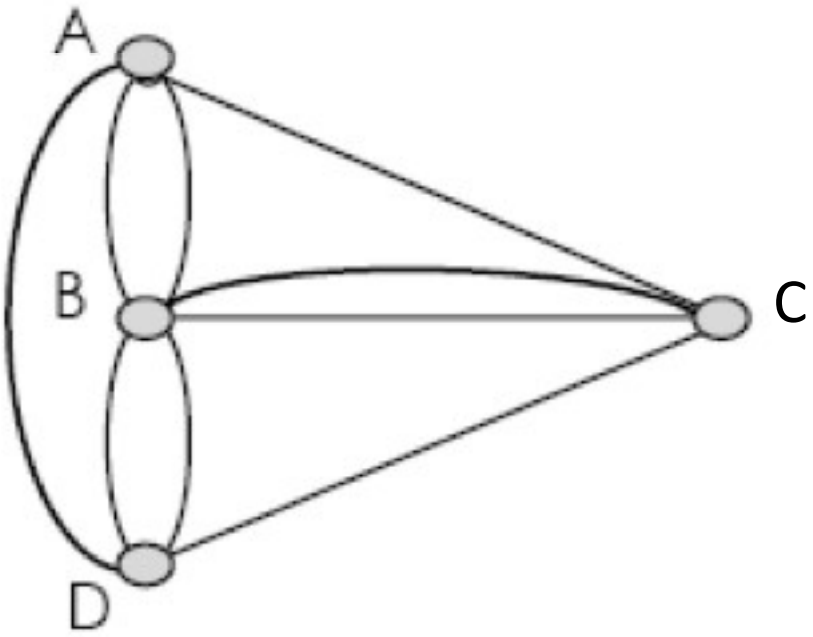


Problem: Starting at one land area, is it possible to walk across all of the bridges exactly once and return to the starting land area?

- Graph of the Königsberg Bridge Problem



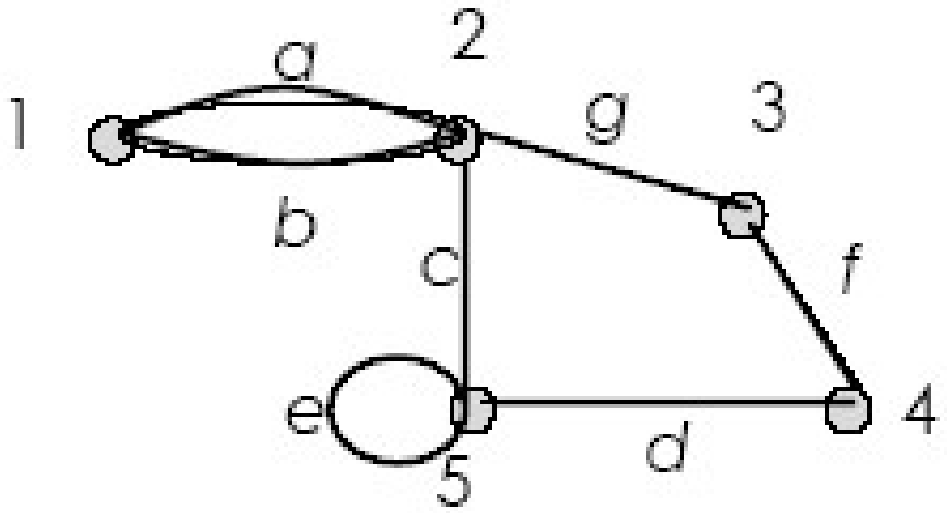
Solution: It is not possible to walk across all of the bridges exactly once and return to the starting land area. Therefore, two additional bridges have been constructed on the Pregel river.



Euler Circuit

Let G be a graph. An Euler circuit for G is a circuit that contains every vertex and every edges of G. That is, an Euler circuit for G is a sequence of adjacent vertices and edges in G that has at least one edges, starts and ends at the same vertex, uses every vertex of G at least once, and uses every edge of G exactly once.

Example 5

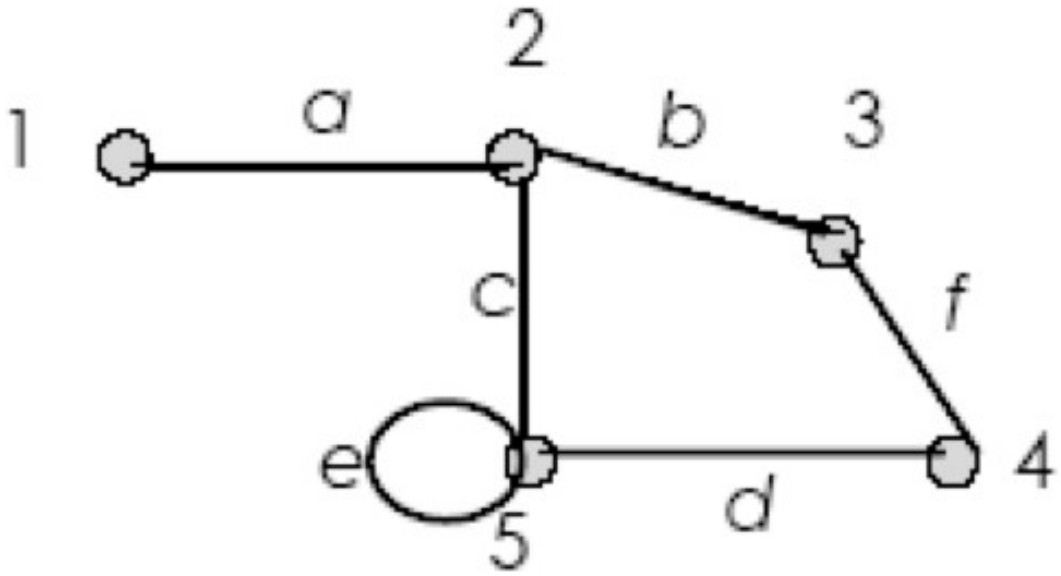


(1, *a*, 2, *c*, 5, *e*, 5, *d*, 4, *f*, 3, *g*, 2, *b*, 1)
is an Euler cycle

Euler Trail

Let G be a graph, and let v and w be two distinct vertices of G. An Euler trail from v to w is a sequence of adjacent vertices and edges that starts at v and ends at w, passes through every **vertex** of G at least once, and traverses every **edge** of G exactly once.

Example 6



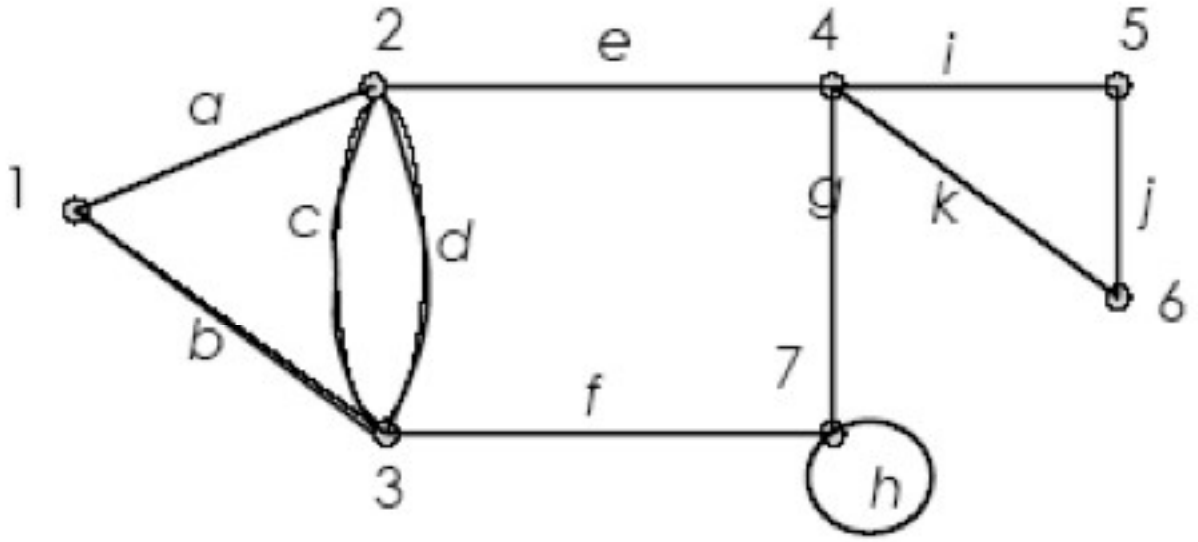
(1, a, 2, c, 5, e, 5, d, 4, f, 3, b, 2) is an Euler trail

Theorem - Euler

- If G is a connected graph and every vertex has even degree, then G has an Euler circuit.

- A graph has an Euler trail from v to w ($v \neq w$) if and only if it is connected and v and w are the only vertices having odd degree.

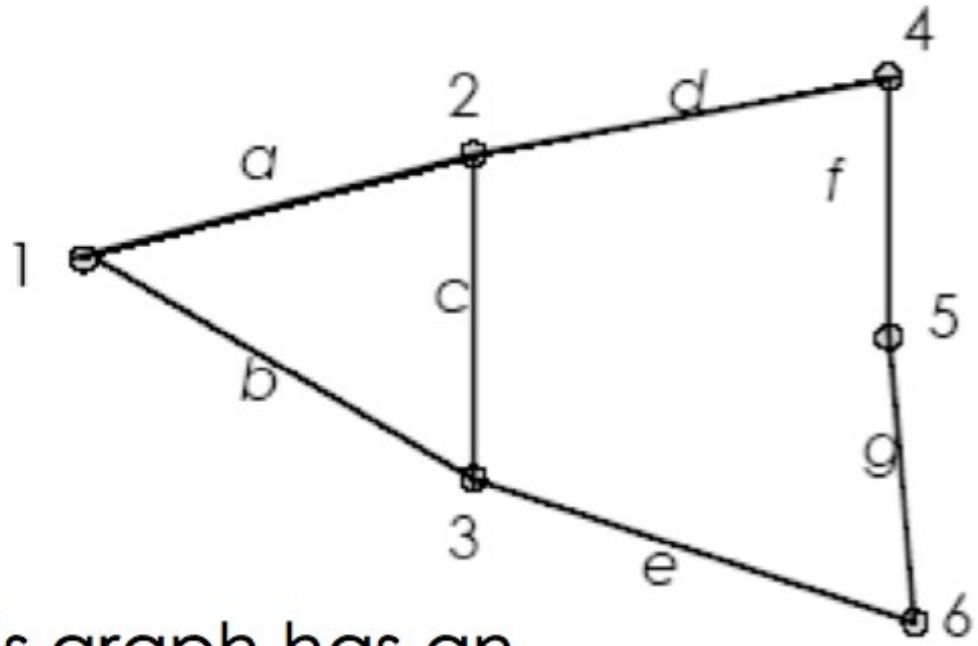
Example 7



This graph has an Euler cycle

Vertex	1	2	3	4	5	6	7
Degree	2	4	4	4	2	2	4

Example 8



This graph has an Euler trail.

Vertex	1	2	3	4	5	6
Degree	2	3	3	2	2	2

Example 9
Past Year 2015/2016

Determine whether the graph in Figure 3 has an Euler cycle or Euler path. If the graph has an Euler cycle or Euler path, exhibit one; otherwise, give an argument that shows there is no Euler path. (4 marks)

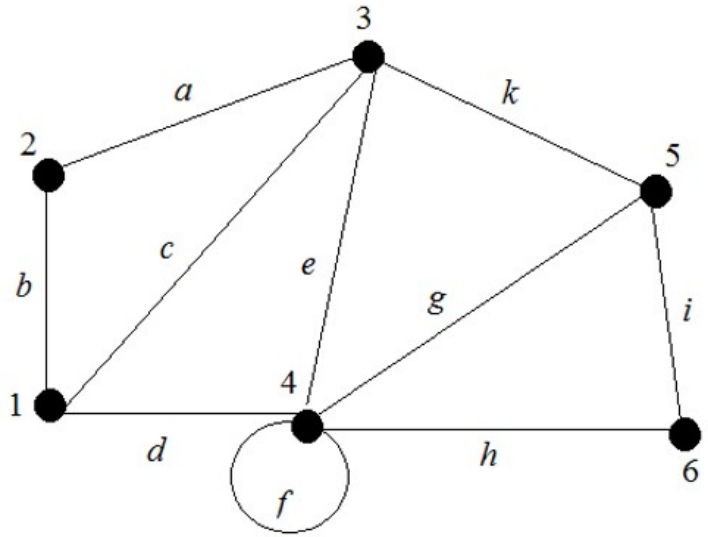


Figure 3

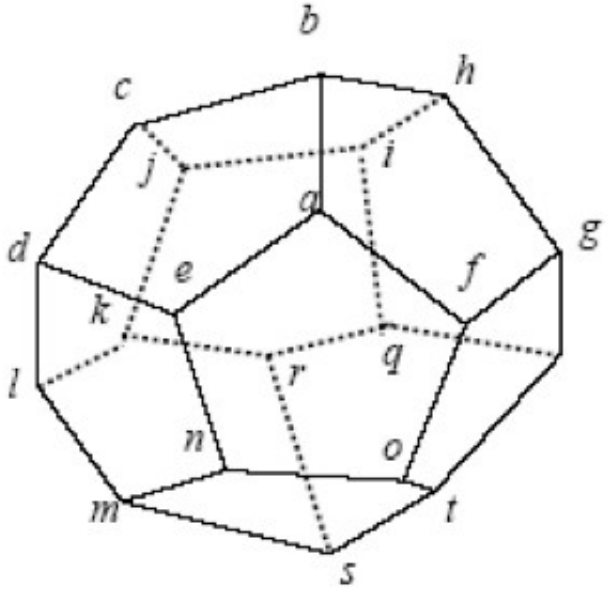
Hamilton Circuits

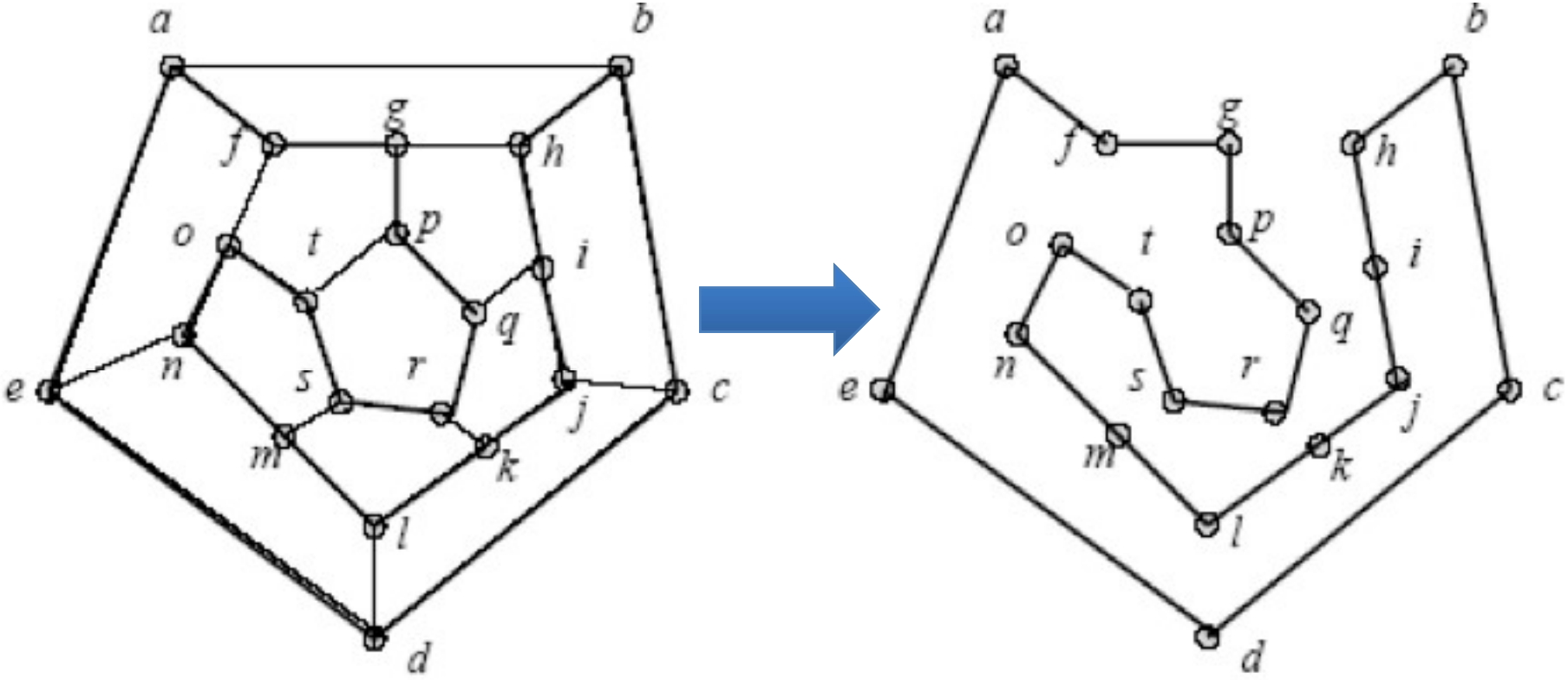
Hamiltonian Circuits

Given a graph G, a Hamiltonian circuit for G is a simple circuit that includes every vertex of G (but doesn't need to include all edges). That is, a Hamiltonian circuit for G is a sequence of adjacent vertices and distinct edges in which every vertex of G appears exactly once, except for the first and the last, which are the same.

Example

- Sir William Rowan Hamilton marketed a puzzle in the mid-1800s in the form of dedocahedron.
- Each corner bore the name of a city.
- The problem was to start at any city, travel along the edges, visit each city exactly one time and return to the initial city.

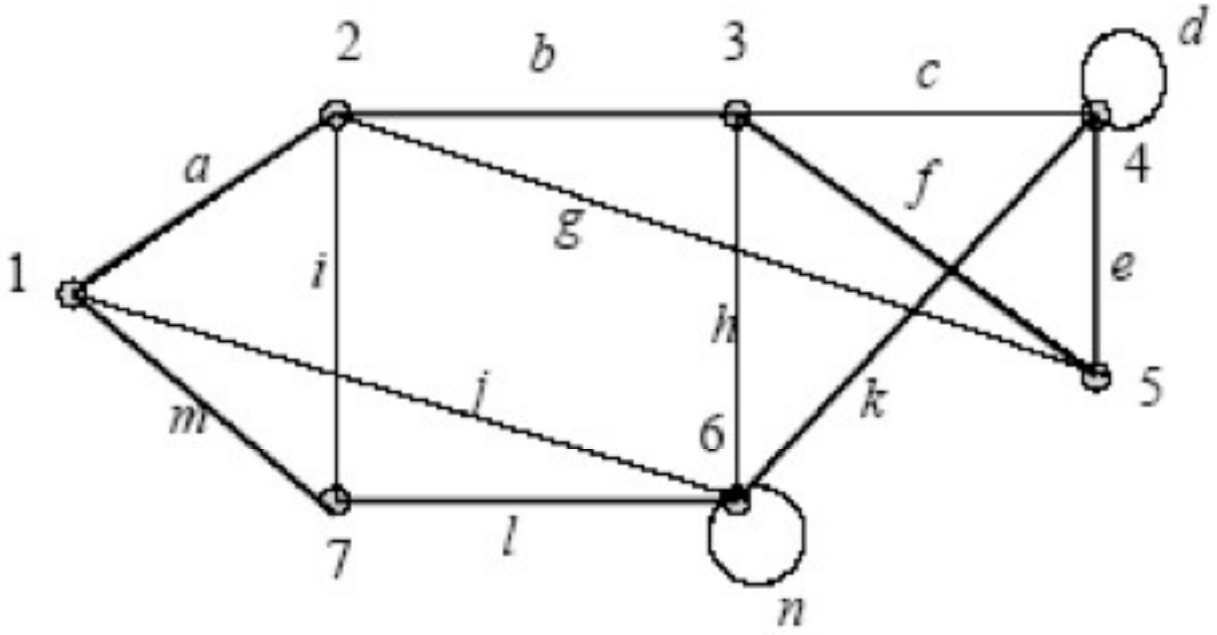




(a): The graph

(b): Hamilton circuit

Example 10



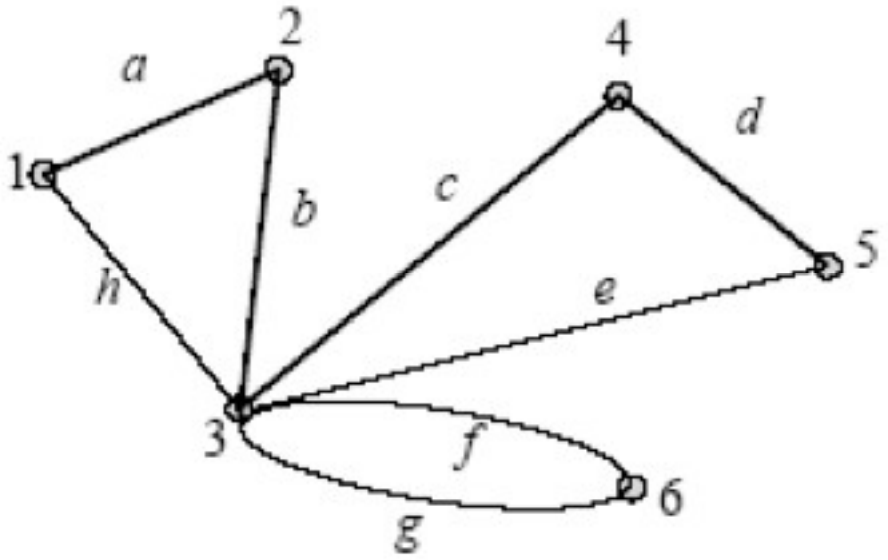
This graph has a Hamilton circuit.

(1, a, 2, b, 3, f, 5, e, 4, k, 6, l, 7, m, 1)

- Visit each vertex just once.

Example 11

This graph does not contain Hamilton circuit.

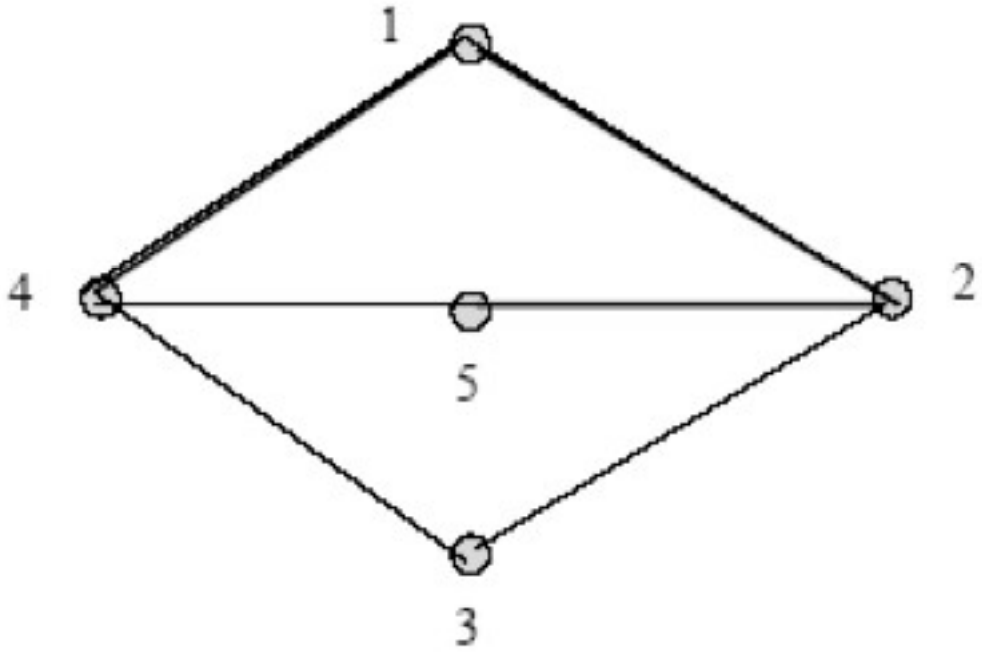


(1, a, 2, b, 3, g, 6, f, 3, e, 5, d, 4, c, 3, h, 1)

- Vertex (3) has to be visited more than once.

Example 12

Question: Is this graph has Hamiltonian cycle?

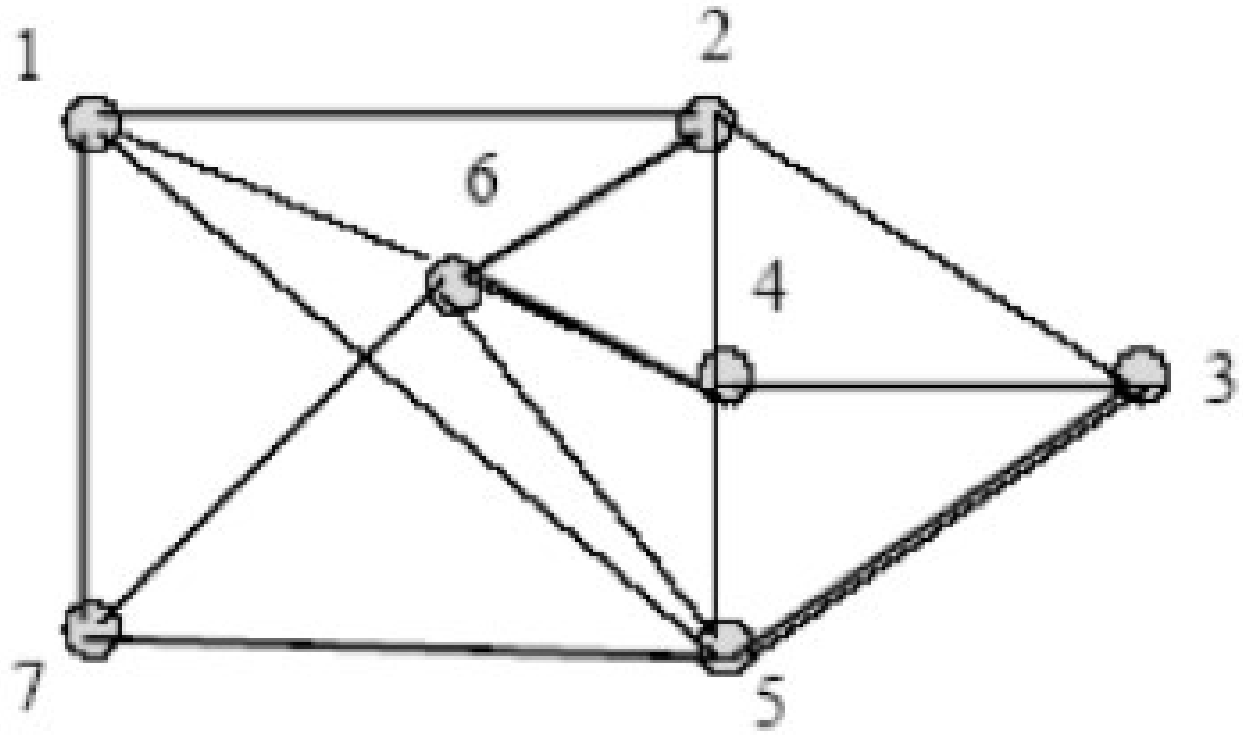


Solution: (1, 4, 3, 2, 5, 4, 1)

- Vertex (4) has to be visited more than once.

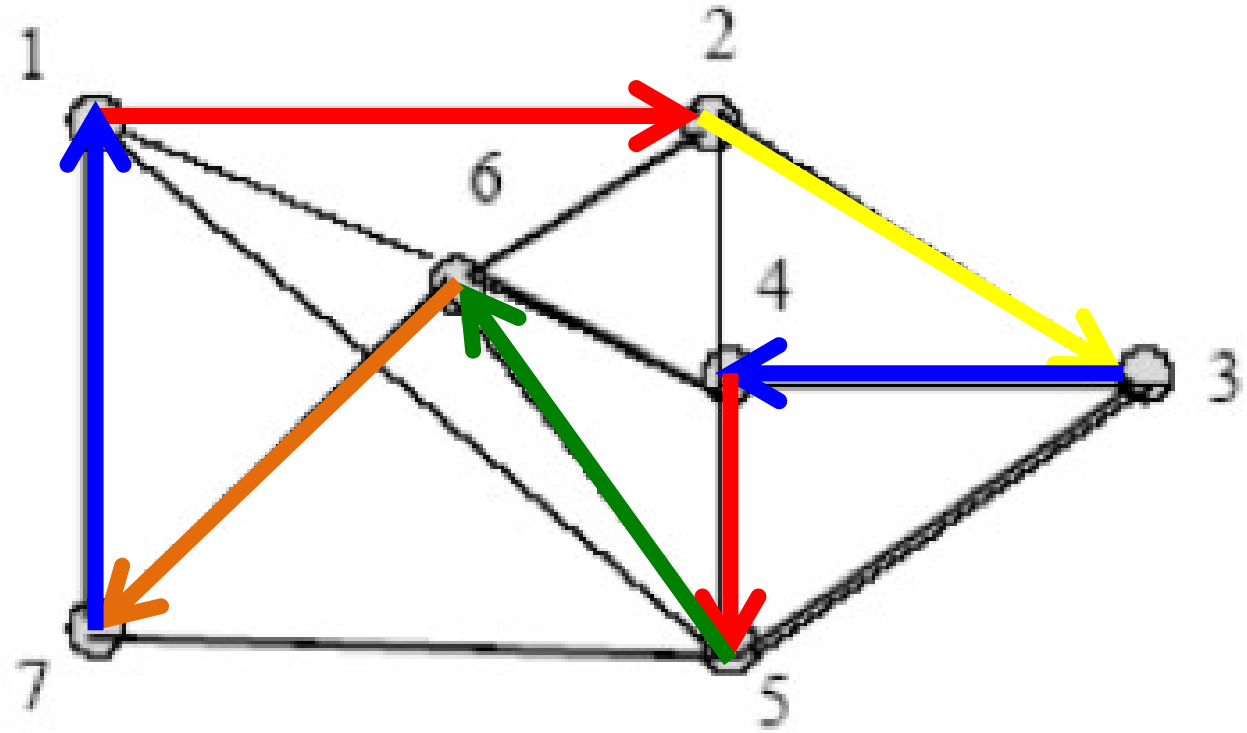
Example 13

Question: Is this graph has Hamiltonian cycle?



Solution

Manually check: This graph has Hamiltonian cycle.



Each vertex has to be visited only once.
For example: (1, 2, 3, 4, 5, 6, 7, 1)

QUESTION 3

An art museum arrange its current exhibit in the seven rooms(Room1,Room2, Room3, Room4,Room5,Room6 and Room 7). Each room is connected with other room by a door (*a*,*b*,*c*,*d*,*e*,*f*,*g*,*h*,*i* and *j*)and the floor plan for the museum is shown in Figure 3.

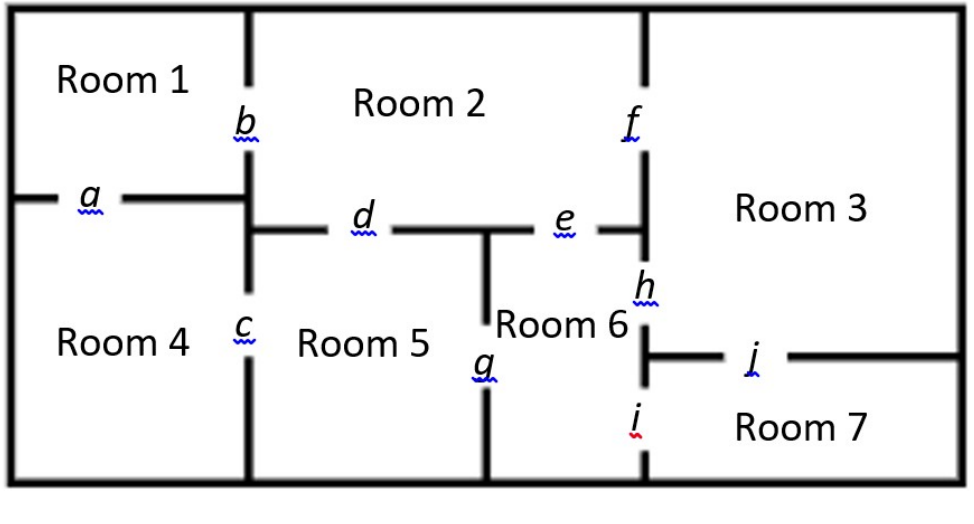


Figure 3: Floor plan for art museum

cont

i) Ramli wants to give a tour at the museum. Is there is a way to tour the exhibit by passing through each door exactly once? Explain. Determine which rooms must they begin and end the tour? Construct the trail of the tour.

ii) Is it possible to tour the museum by visiting each room exactly once (not necessarily using every door)?If possible, construct the tour that begin and end at Room 1.